Interpretable Machine Learning

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BIOS 735: Machine Learning for Public Health Applications

What is Interpretable Machine Learning?

- One way to define Interpretable Machine Learning (IML) is the degree to which a human can understand the cause of a decision made by a machine learning model.
- It's a critical aspect, especially in fields where ML is being used to determine decisions that can affect human lives.... why?

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- Interpretability is not required if the model has no significant impact.

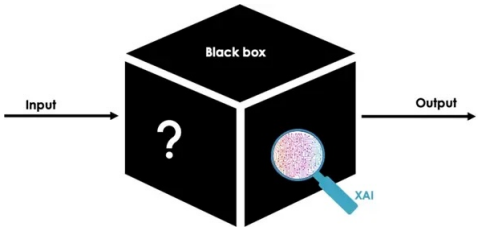
"Imagine someone named Mike working on a machine learning side project to predict where his friends will go for their next holidays based on Facebook data. Mike just likes to surprise his friends with educated guesses where they will be going on holidays."

IML versus XAI

- There a whole book titled *Interpretable Machine Learning*, however, this book is 2/3rd's about what we will call Explainable AI (XAI) and 1/3rd about IML.
- What do you think the difference between IML and XAI are?

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Healthcare Diagnosis

- Machine learning models are increasingly used to diagnose diseases from medical images, like MRI scans or pathology slides. Interpretability is essential here to:
  - Understand the model's reasoning for its diagnosis.
  - Gain trust from medical professionals who will act on these diagnoses.
  - Identify if the model is using sensible medical indicators or if it's being misled by irrelevant artifacts in the data.

Personalized Medicine

- In personalized medicine, interpretability helps in:
  - Understanding why certain treatments are recommended based on a patient's genetic makeup.
  - Ensuring that it is actually genetics that are driving the results.
  - Allowing doctors to combine machine-generated insights with their clinical expertise.
  - Ensuring that patients receive treatments that are effective for their specific conditions.

IVF

- In modern IVF, black box models that have not been subjected to randomized controlled trials determine which embryos to transfer to the woman.
- AI is used as an embryo selection tool to improve the success rate per transfer.
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- It could lead to:
  - Misrepresentation of Patient Values (maybe the model certain traits)
  - Health and Well-Being of Future Children
  - Impacts of Disvaluing Diversity (e.g., disabilities)

COMPAS

- Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) provides a black box proprietary algorithm for predicting criminal recidivism.
- The COMPAS algorithm has been accused of being racially biased (link), though this has never been proven.
- Wang et al. (2020) compared the COMPAS model with a very simple decision tree and found that it predicted recidivism more accurately.

Interpretable Machine Learning

- Here's a definition of IML from Rudin et al. (2021):

  "An interpretable machine learning model obeys a domain-specific set of constraints to allow it (or its predictions, or the data) to be more easily understood by humans. These constraints can differ dramatically depending on the domain."

- A typical interpretable supervised learning setup, with data $\{z_i\}_i$, and models chosen from function class $\mathcal{F}$ is:

$$\min_{f \in \mathcal{F}} \sum_i L(f, z_i) + C \cdot \text{InterpretabilityPenalty}(f)$$

  such that f is within an interpretability constraint.

Interpretable Machine Learning

- These constraints aim to make the resulting model f – or its predictions – more interpretable.
- The interpretability penalties or constraints depend on the model and application.
  - For most regression/classification: sparsity of the model, monotonicity with respect to a variable,
  - Computer vision: case-based reasoning or disentanglement, and visual understanding of intermediate computations
  - Generative constraints (e.g., laws of physics)

Interpretable Machine Learning

- Creating interpretable models can sometimes be much more difficult than creating black box models for many different reasons
  1. Solving the optimization problem may be computationally hard, depending on the choice of constraints and the model class $\mathcal{F}$.
  2. When one does create an interpretable model, one invariably realizes that the data are problematic and require troubleshooting, which slows down deployment (but leads to a better model).
  3. It might not be initially clear which definition of interpretability to use.

Interpretability and accuracy

- Rudin et al (2021) state that:

  "It is important not to assume that one needs to make a sacrifice in accuracy in order to gain interpretability. In fact, interpretability often begets accuracy and not the reverse. Interpretability versus accuracy is, in general, a false dichotomy in machine learning."

- **One clear example:** sparse linear models can be more accurate than dense linear models.

Interpretability and the scientific process

- Knowledge Discovery process
- How can we tune the processing of the data, the loss function, or the evaluation metric if we can't understand how the model works?



Interpretabe ML Models

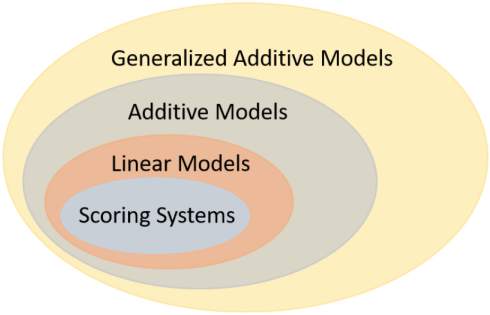
- **Transparent Models:** These are models that are inherently interpretable.
- Examples include linear regression, logistic regression, or decision trees.
- Their decision-making process is relatively straightforward and can be followed by humans.
- Nearest-neighbor methods are also relatively transparent.

Scoring Systems

Patient screens positive for obstructive sleep apnea if Score >1			
1.	age ≥ 60	4 points	.....
2.	hypertension	4 points	+ .....
3.	body mass index ≥ 30	2 points	+ .....
4.	body mass index ≥ 40	2 points	+ .....
5.	female	-6 points	+ .....
	Add points from row 1-6	Score	= .....

Table 2: A scoring system for sleep apnea screening (Ustun et al., 2016). Patients that screen positive may need to come to the clinic to be tested.

Linear models and extensions



$$g\{E(Y)\} = f(X_1) + f(X_2) + \ldots + f(X_P)$$

Introduction

- *Explainable AI (XAI)* or *Post-hoc Interpretability*, is where one attempts to explain a black box using an approximation model, derivatives, variable importance measures, or other statistics.
- To some, XAI is a pointless endeavor as it doesn't adhere to the initial definition of interpretable ML we first discussed.

  "An interpretable machine learning model obeys a domain-specific set of constraints to allow it (or its predictions, or the data) to be more easily understood by humans. These constraints can differ dramatically depending on the domain."

- However, we'll review some of these approaches (we've many of these already).

Types of Interpretability

Global vs. Local

- **Global:** How predictors influence predictions across the entire dataset (e.g., variable importance, partial dependence).
- **Local:** Why the model made a specific prediction for one case (e.g., SHAP, LIME).

Model-Specific vs. Model-Agnostic

- **Model-specific:** Built into the model (e.g., coefficients in linear models).
- **Model-agnostic:** Applicable to any model (e.g., variable importance, SHAP).

Why Interpretability Matters in Public Health

- Enhances trust and transparency in predictive models.
- Aids in identifying modifiable risk factors.
- Ensures fairness, equity, and accountability in decision-making.
- Helps bridge the gap between data science and domain expertise.

Variable Importance (VIP)

- **Goal:** Quantify how much each predictor contributes to model performance.
- **Model-based definitions:**
  - For linear models, importance corresponds to $|\beta_j|$ or $|\beta_j| \times \text{SD}(X_j)$.
  - For tree-based models, variable importance is often based on total reduction in impurity (e.g., Gini or deviance).
- **Permutation importance:** A model-agnostic approach — measure the drop in predictive accuracy when variable X_j is randomly permuted:

$$I_j = \text{Accuracy}_{\text{original}} - \text{Accuracy}_{\text{permuted}(X_j)}.$$

- **Interpretation:** Larger I_j indicates the model relied more heavily on variable X_j.

Partial Dependence Plots (PDPs)

- **Purpose:** Describe the marginal relationship between a predictor and model predictions.
- For a fitted model $\hat{f}(X)$, the partial dependence function for feature X_j is:

$$PD_j(x_j) = \frac{1}{n} \sum_{i=1}^{n} \hat{f}(x_j, x_{i,-j})$$

  where $x_{i,-j}$ are all other features for observation i.
- The curve of $PD_j(x_j)$ shows how the expected prediction changes with X_j.
- **Interpretation:** Nonlinear or non-monotone shapes reveal complex effects such as thresholds or plateaus.
- Caution: Assumes independence between X_j and X_{-j}.

Accumulated Local Effects (ALE)

- ALE addresses the correlation problem in PDPs by examining *local* effects.
- Defined as the accumulated average change in predictions when X_j increases from z_{k-1} to z_k:

$$ALE_j(z_k) = \sum_{l=1}^{k} \frac{1}{n_l} \sum_{i:z_{l-1}<x_{ij}\leq z_l} [\hat{f}(x_{ij} = z_l, x_{i,-j}) - \hat{f}(x_{ij} = z_{l-1}, x_{i,-j})]$$

- ALE plots are centered so that the mean effect is zero.
- **Interpretation:** Displays the *local slope* of the model's prediction surface, aggregated to show the net effect of X_j.

SHAP Values

- Based on Shapley values from cooperative game theory.
- Each feature receives a fair share of the total prediction difference from the model's baseline:

$$\phi_j = \sum_{S \subseteq \{1,\ldots,p\} \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} [\hat{f}(S \cup \{j\}) - \hat{f}(S)]$$

- **Interpretation:** ϕ_j measures the average contribution of feature j to the prediction across all possible subsets of features.
- SHAP values provide both **direction** (positive or negative influence) and **magnitude** (strength of influence).
- **Benefits:** Consistent across models, locally additive, and theoretically grounded.

Local Surrogate Models and LIME

- **Idea:** Approximate a complex model locally with a simple, interpretable model.
- **Method:**
  1. Choose an observation of interest x_0.
  2. Generate perturbed samples around x_0.
  3. Weight them by proximity to x_0.
  4. Fit a simple model (e.g., linear regression) on the weighted data to approximate $\hat{f}$ near x_0.
- **Result:** The coefficients of the local surrogate describe which variables are most influential for that prediction.
- **Interpretation:** Explains the reasoning for a single prediction; helpful for communicating individual-level risk in health applications.

Comparing Interpretability Methods

Method	**Type**	**Strengths / Limitations**
Variable Importance	Global	Simple, quantitative, but lacks directionality
PDP / ALE	Global	Visual, can reveal nonlinear effects; may assume independence
SHAP	Both	Theoretically consistent; computationally intensive
LIME	Local	Easy to interpret, but sensitive to sampling choices

Communicating Interpretability

- Visualizations should focus on clarity, not aesthetics.
- Summarize model findings in terms of public-health meaning.
- Always report uncertainty (e.g., variability in SHAP or ALE estimates).
- Use plain language when communicating with non-technical audiences.

Ethical and legal considerations

- The ability to interpret a model is closely linked to stakeholders' trust in the model's decisions.
- Interpretable models do not necessarily create or enable trust – they could also enable distrust. They simply allow users to decide whether to trust them.
- Interpretability is especially important in areas with ethical implications, such as criminal justice, loan approval, or patient treatment.
- Interpretability is also driven by legal requirements, such as the General Data Protection Regulation (GDPR, an EU law) "*right to explanation*," which mandates that users have the right to be given explanations for algorithmic decisions that affect them.

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